

Lecture 2 (Additional Notes)

Recall these properties of $\mathbb{Z}$:

1. $\mathbb{Z}$ is a subring of $\mathbb{Q}$.
2. Its field of fractions is $\mathbb{Q}$. That is $\text{ff}(\mathbb{Z}) = \mathbb{Q}$.
3. (Gauss's Lemma) If $f(x) \in \mathbb{Z}[x]$ is monic and if $b \in \mathbb{Q}$ is a root of f , then $b \in \mathbb{Z}$.
4. $\mathbb{Z}$ is the smallest subring of $\mathbb{Q}$ for which 2. and 3. hold.
5. Every nonzero $b \in \mathbb{Z}$ can be written as a product of prime powers, and this factorization is unique up to ordering.
6. If $p \in \mathbb{Z}$ is a prime, then $\mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p$ is a field.

The ring of integers of L

When $L/\mathbb{Q}$ is a finite extension, we would like to find a subring that recovers as many of the properties above as possible. The subset of L that does this is

$$\mathcal{O}_L = \{\alpha \in L \mid m_{\alpha, \mathbb{Q}}(x) \in \mathbb{Z}[x]\}.$$

We call $\mathcal{O}_L$ the ring of integers of L . It requires proof that this is a subring of L , but it is, and it is the finitely generated abelian group within L that best serves as an analogy to $\mathbb{Z} \subset \mathbb{Q}$.

It is not obvious that $\mathcal{O}_L \subset L$ satisfies properties 2. through 4., but it does, and we will use those properties as needed. Some examples of extensions $L/\mathbb{Q}$ and their corresponding rings of integers are given below:

L	$\mathcal{O}_L$
$\mathbb{Q}$	$\mathbb{Z}$
$\mathbb{Q}(i)$	$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$
$\mathbb{Q}(\sqrt{-5})$	$\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} \mid a, b \in \mathbb{Z}\}$
$\mathbb{Q}(\sqrt{13})$	$\mathbb{Z}\left[\frac{1+\sqrt{13}}{2}\right] = \{a + b\left(\frac{1+\sqrt{13}}{2}\right) \mid a, b \in \mathbb{Z}\}$
$\mathbb{Q}(\zeta_n)$	$\mathbb{Z}[\zeta_n]$

(There is an Exercise in the problem set: Find an $\mathcal{O}_L$ for which $\mathcal{O}_L \neq \mathbb{Z}[\beta]$ for any $\beta \in L$.)

Property 5. is the first of the properties that can fail. There are finite extensions $L/\mathbb{Q}$ for which $\mathcal{O}_L$ also has unique factorization of nonzero elements, for example $\mathbb{Q}(i)$. But there are extensions for which you cannot uniquely factor every nonzero element of $\mathcal{O}_L$, for example $\mathbb{Q}(\sqrt{-5})$: you can factor 6 in two distinct ways: $6 = 2(3) = (1 + \sqrt{-5})(1 - \sqrt{-5})$. In fact, there are infinitely many $L/\mathbb{Q}$ for which property 5. fails. (See, for example, Gauss's Class Number One conjecture, and the failure of proofs to use $\mathbb{Z}[\zeta_p]$ to prove Fermat's Last Theorem.) Although property 5 fails for factoring elements of $\mathcal{O}_L$, an appropriate re-imagining allows us to recover a different kind of unique factorization. Rather than factoring elements of $\mathcal{O}_L$, we can factor ideals of $\mathcal{O}_L$, and then we do have unique factorization of nonzero ideals.

Definition (Ideal): If R is a commutative ring (e.g., $\mathcal{O}_L$), then I is an ideal of R if it is a subgroup of R under addition and if for every $a \in I$, and every $b \in R$, $ab \in I$.

You have seen ideals before: If $n \in \mathbb{Z}$, then the set of all multiples of n is an ideal of $\mathbb{Z}$.

In order to recover something close to unique factorization, we need to generalize idea of what “prime” means.

Definition (Prime Ideal): An ideal $\mathfrak{p}$ of R is a prime ideal if for any $a, b \in R$, if $ab \in \mathfrak{p}$ then either $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.

Examples: If $p \in \mathbb{Z}$ is a prime number, then $p\mathbb{Z} = \{pn \mid n \in \mathbb{Z}\}$ is a prime ideal. The ideal $\{0\}$ is also a prime ideal of $\mathbb{Z}$.

The property of $\mathcal{O}_L$ that we can use to (as much as possible) compensate for the failure of property 5. is the following:

Theorem: If $A \subseteq \mathcal{O}_L$ is a nonzero ideal, then A can be factored as

$$A = \mathfrak{p}_1^{e_1} \mathfrak{p}_2^{e_2} \cdots \mathfrak{p}_r^{e_r},$$

where each $\mathfrak{p}_i$ is a prime ideal of $\mathcal{O}_L$. This factorization is unique up to ordering.

With this new notion of prime, we can also recover a property similar to 6.:

Theorem: If $\mathfrak{p}$ is a nonzero prime ideal of $\mathcal{O}_L$, then the quotient $\mathcal{O}_L/\mathfrak{p}$ is a finite field.

There are more important results we'll need to use about these prime ideals $\mathfrak{P}$ of $\mathcal{O}_L$, but for now, two that you should know are that

(1) There is a unique prime $p \in \mathbb{Z}$ for which $\mathfrak{P} \cap \mathbb{Z} = p\mathbb{Z}$. More generally, if $\mathbb{Q} \subseteq K \subseteq L$ are finite extensions, then for a prime $\mathfrak{P}$ of $\mathcal{O}_L$, there is a unique prime $\mathfrak{p}$ of $\mathcal{O}_K$ such that $\mathfrak{P} \cap K = \mathfrak{p}$.

(2) $\mathcal{O}_L/\mathfrak{P}$ is a finite degree extension of $\mathbb{F}_p$. More generally, if $\mathbb{Q} \subseteq K \subseteq L$ are finite extensions, then for a prime $\mathfrak{P}$ of $\mathcal{O}_L$ and the unique prime $\mathfrak{p}$ of $\mathcal{O}_K$ such that $\mathfrak{P} \cap K = \mathfrak{p}$, $\mathcal{O}_L/\mathfrak{P}$ is a finite degree extension of the finite field $\mathcal{O}_K/\mathfrak{p}$.